

```
!pip install ultralytics opencv-python matplotlib
```

Collecting ultralytics

Downloading ultralytics-8.4.163-py3-none-any.whl.metadata (46 kB)

46.7/46.7 kB 2.5 MB/s eta 0:00:00

Requirement already satisfied: opencv-python in /usr/local/lib/python3.13/dist-packages (5.0.0.93)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)

Requirement already satisfied: cloudpickle>=3.1.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (3.1.2)

Requirement already satisfied: filelock>=3.16.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (3.32.5)

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.1.3)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (11.3.0)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.32.4)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.11.0+cpu)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (0.26.0+cpu)

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (1.35.2)

Collecting nvidia-ml-py>=12.0.0 (from ultralytics)

Downloading nvidia_ml_py-13.615.71-py3-none-any.whl.metadata (9.7 kB)

Collecting ultralytics-thop>=2.1.6 (from ultralytics)

Downloading ultralytics_thop-2.1.6-py3-none-any.whl.metadata (13 kB)

Collecting ultralytics-platform>=0.1.45 (from ultralytics)

Downloading ultralytics_platform-0.1.65-py3-none-any.whl.metadata (16 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (4.64.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (26.3)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.13/dist-packages (from polars>=0.20.0->ultral

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultra

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultral

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultral

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests>=2.23.0->ultral

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultral

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultral

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultral

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultral

Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultral

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultral

Requirement already satisfied: httpx<1,>=0.28 in /usr/local/lib/python3.13/dist-packages (from ultralytics-platform>=0.1.45->ult

Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28->ultral

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.28->ultral

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.28->ultral

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch>=1.8.0->

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->torch>=1.8.0->ultral

Downloading ultralytics-8.4.163-py3-none-any.whl (1.5 MB)

1.5/1.5 MB 22.9 MB/s eta 0:00:00

Downloading nvidia_ml_py-13.615.71-py3-none-any.whl (58 kB)

58.1/58.1 kB 5.2 MB/s eta 0:00:00

Downloading ultralytics_platform-0.1.65-py3-none-any.whl (91 kB)

91.0/91.0 kB 6.5 MB/s eta 0:00:00

Downloading ultralytics_thop-2.1.6-py3-none-any.whl (30 kB)

Installing collected packages: nvidia-ml-py, ultralytics-thop, ultralytics-platform, ultralytics

Successfully installed nvidia-ml-py-13.615.71 ultralytics-8.4.163 ultralytics-platform-0.1.65 ultralytics-thop-2.1.6

```
from ultralytics import YOLO
import cv2
import matplotlib.pyplot as plt
```

```
print("Libraries imported successfully!")
```

Creating new Ultralytics Settings v0.0.8 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics>

Libraries imported successfully!

```
model = YOLO("yolov8n.pt")
```

```
print("YOLOv8 model loaded successfully!")
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt': 100%  6.2MB 7

YOLOv8 model loaded successfully!

```
results = model("birds.jpg")

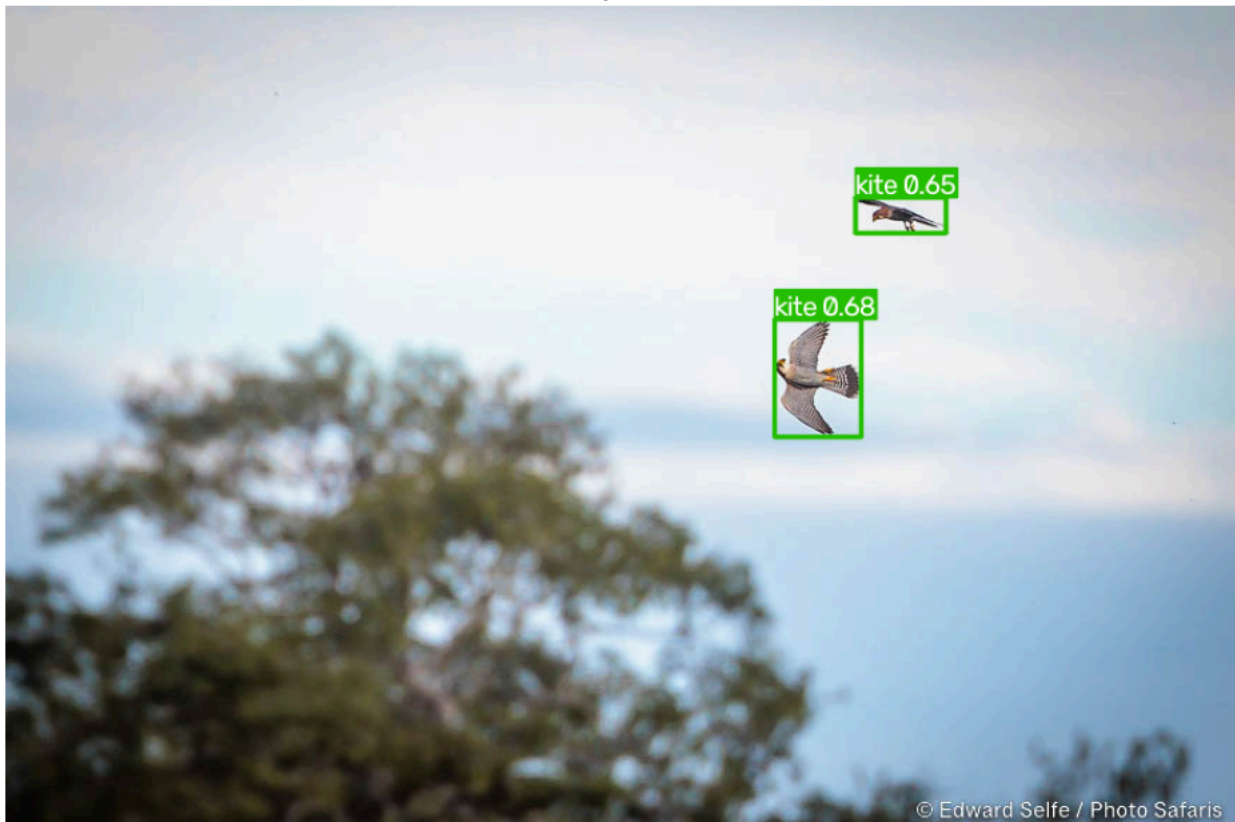
print("Object detection completed!")
```

image 1/1 /content/birds.jpg: 448x640 2 kites, 396.2ms
 Speed: 29.2ms preprocess, 396.2ms inference, 34.1ms postprocess per image at shape (1, 3, 448, 640)
 Object detection completed!

```
result_image = results[0].plot()

plt.figure(figsize=(12, 8))
plt.imshow(cv2.cvtColor(result_image, cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.title("YOLOv8 Object Detection")
plt.show()
```

YOLOv8 Object Detection



```
for result in results:
    boxes = result.boxes

    for box in boxes:
        class_id = int(box.cls[0])
        confidence = float(box.conf[0])

        class_name = model.names[class_id]

        print(
            f"Object: {class_name} | "
            f"Confidence: {confidence:.2f}"
        )
```

```
Object: kite | Confidence: 0.68
Object: kite | Confidence: 0.65
```

```
total_objects = 0

for result in results:
    total_objects += len(result.boxes)

print("Total objects detected:", total_objects)
```

Total objects detected: 2

```
results[0].save(filename="detected_output.jpg")
```

```
print("Detected image saved successfully!")
```

Detected image saved successfully!

```
from IPython.display import Image, display
```

```
display(Image(filename="detected_output.jpg"))
```

